Flow Matching

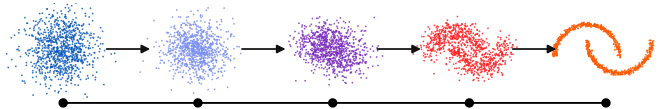
Learn velocity field that transports p_0 to p_1



$$v^*(\mathcal{U}_\tau, \tau) = \tau(\mathcal{U}_1 - \mathcal{U}_0) + \mathcal{U}_0$$

Stochastic Interpolant Process

$$\mathcal{U}_\tau = \mathcal{I}(\tau, \mathcal{U}_0, \mathcal{U}_1) + \gamma(\tau)\,\epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$



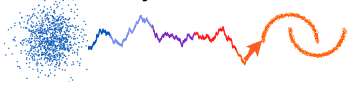
$\tau = 0$ $\tau = 0.25$ $\tau = 0.5$ $\tau = 0.75$ $\tau = 1$

$\mathcal{U}_0 \sim p_0$ $\mathcal{U}_1 \sim p_1$

Marginal distribution $p_\tau, \tau \in [0, 1]$

Score-Based Generative Models

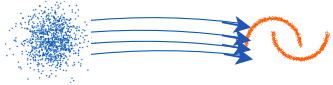
Use score $\nabla_{\mathcal{U}} \log p_\tau(\mathcal{U}_\tau)$ to define dynamics



$$d\mathcal{U}_\tau = [f_\tau(\mathcal{U}_\tau) - \frac{1}{2} g_\tau^2 \nabla_{\mathcal{U}} \log p_\tau(\mathcal{U}_\tau) d\tau + g_\tau dW_\tau$$

Continuous Normalizing Flows

deterministic, invertible probability flow w/o noise



$$d\mathcal{U}_\tau / d\tau = v_\tau(\mathcal{U}_\tau, \tau)$$

Diffusion Models

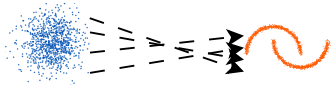
SDE with fixed forward noise schedule



$$d\mathcal{U}_\tau = f_\tau(\mathcal{U}_\tau) d\tau + g_\tau dW_\tau$$

Entropy-Regularized Optimal Transport

Transport mass between p_0 and p_1 with controlled stochasticity $\epsilon > 0$



Schrödinger Bridges

Most likely stochastic process between p_0 and p_1



$$d\mathcal{U}_\tau = [f_\tau(\mathcal{U}_\tau) + g_\tau^2 \nabla_{\mathcal{U}} \log \varphi_\tau(\mathcal{U}_\tau)] d\tau + g_\tau dW_\tau$$